

Project name : “Individual_Assignment_4”

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Course: HDS 5310

Project Purpose: To examine how technology use varies by sex and marital status using GSS 2018 data through descriptive analysis and ANOVA.

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STEP 1 : Load File

```
#Load GSS rda file and Assign GSS to gss.2018
load(file="C:/Users/drhud/Downloads/Individual Assignment 4/Individual Assignment 4/gss
2018.rda")
```

```
gss.2018<-GSS
```

```
#Removing GSS
rm(GSS)
```

```
#Display summary
summary(object = gss.2018)
```

```
##      YEAR      BALLOT      USETECH      HAPPY
## Min.   :2018   Min.   :1.000   Min.   : -1.00   Min.   :1.000
## 1st Qu.:2018   1st Qu.:1.000   1st Qu.: -1.00   1st Qu.:1.000
## Median :2018   Median :2.000   Median : 10.00   Median :2.000
## Mean   :2018   Mean   :2.002   Mean   : 48.09   Mean   :1.855
## 3rd Qu.:2018   3rd Qu.:3.000   3rd Qu.: 80.00   3rd Qu.:2.000
## Max.   :2018   Max.   :3.000   Max.   :999.00   Max.   :8.000
## PARTYID      RINCOME      RACE      SEX
## Min.   :0.000   Min.   : 0.000   Min.   :1.000   Min.   :1.000
## 1st Qu.:1.000   1st Qu.: 0.000   1st Qu.:1.000   1st Qu.:1.000
## Median :3.000   Median : 9.000   Median :1.000   Median :2.000
## Mean   :2.968   Mean   : 7.509   Mean   :1.394   Mean   :1.552
## 3rd Qu.:5.000   3rd Qu.:12.000   3rd Qu.:2.000   3rd Qu.:2.000
```

```
## Max. :9.000 Max. :98.000 Max. :3.000 Max. :2.000
## DEGREE EDUC AGE MARITAL
## Min. :0.000 Min. : 0.00 Min. :18.00 Min. :1.00
## 1st Qu.:1.000 1st Qu.:12.00 1st Qu.:34.00 1st Qu.:1.00
## Median :1.000 Median :14.00 Median :48.00 Median :2.00
## Mean :1.684 Mean :13.84 Mean :49.13 Mean :2.67
## 3rd Qu.:3.000 3rd Qu.:16.00 3rd Qu.:63.00 3rd Qu.:5.00
## Max. :4.000 Max. :99.00 Max. :99.00 Max. :9.00
## HRS2 HRS1 WRKSTAT ID_
## Min. :-1.00000 Min. :-1.00 Min. :1.000 Min. : 1
## 1st Qu.: -1.00000 1st Qu.: -1.00 1st Qu.:1.000 1st Qu.: 588
## Median : -1.00000 Median :30.00 Median :2.000 Median :1176
## Mean : 0.08017 Mean :24.47 Mean :2.963 Mean :1175
## 3rd Qu.: -1.00000 3rd Qu.:40.00 3rd Qu.:5.000 3rd Qu.:1762
## Max. :99.00000 Max. :99.00 Max. :9.000 Max. :2348
## UNHAPPY
## Min. :0.000
## 1st Qu.:0.000
## Median :0.000
## Mean :1.039
## 3rd Qu.:2.000
## Max. :9.000
```

```
#loading library
library(tidyverse)
```

```
## — Attaching core tidyverse packages ————— tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2     3.5.1      ✓ tibble     3.2.1
## ✓ lubridate  1.9.4      ✓ tidyr      1.3.1
## ✓ purrr      1.0.2
## — Conflicts ————— tidyverse_conflicts() —
## * dplyr::filter() masks stats::filter()
## * dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
```

```
##
##      some
```

STEP 2 : Cleaning the marital status, sex, and tech use variables

```
gss.2018.cleaned <- gss.2018 %>%
  #Select relevant variables and begin cleaning process
  select(SEX, USETECH, MARITAL) %>%
  mutate(USETECH = na_if(USETECH, -1)) %>%
  mutate(USETECH = na_if(USETECH, 998)) %>%
  mutate(USETECH = na_if(USETECH, 999)) %>%
  # Convert SEX to a factor with clear labels
  mutate(SEX = factor(x = SEX, labels = c("male", "female"))) %>%
  # Replace '9' in MARITAL with NA
  mutate(MARITAL = na_if(x = MARITAL, y = 9)) %>%
  # Re-code MARITAL values with descriptive labels
  mutate(MARITAL = factor(MARITAL,
    levels = c(1, 2, 3, 4, 5),
    labels = c("Married",
               "Widowed",
               "Divorced",
               "Separated",
               "Never married"))
  )
```

```
#displaying cleaned summary
summary(object = gss.2018.cleaned)
```

```
##      SEX      USETECH      MARITAL
## male :1051  Min.   : 0.00  Married   :998
## female:1294 1st Qu.: 15.00  Widowed   :200
##           Median : 60.00  Divorced   :402
##           Mean   : 55.15  Separated   : 75
##           3rd Qu.: 90.00  Never married:668
##           Max.   :100.00  NA's       : 2
##           NA's    :936
```

#Interpretation Interpretation: The data set includes more females than males. Tech use ranges from 0 to 100 with a median of 60, and a substantial number of missing values (936). Most participants are married followed by never married.

STEP 3 : Descriptive stats and visualizations

```
#Calculate overall mean and standard deviation of tech use (excluding missing values)
gss.2018.cleaned %>%
  drop_na(USETECH) %>%
  summarize(
    mean_use = mean(USETECH),
```

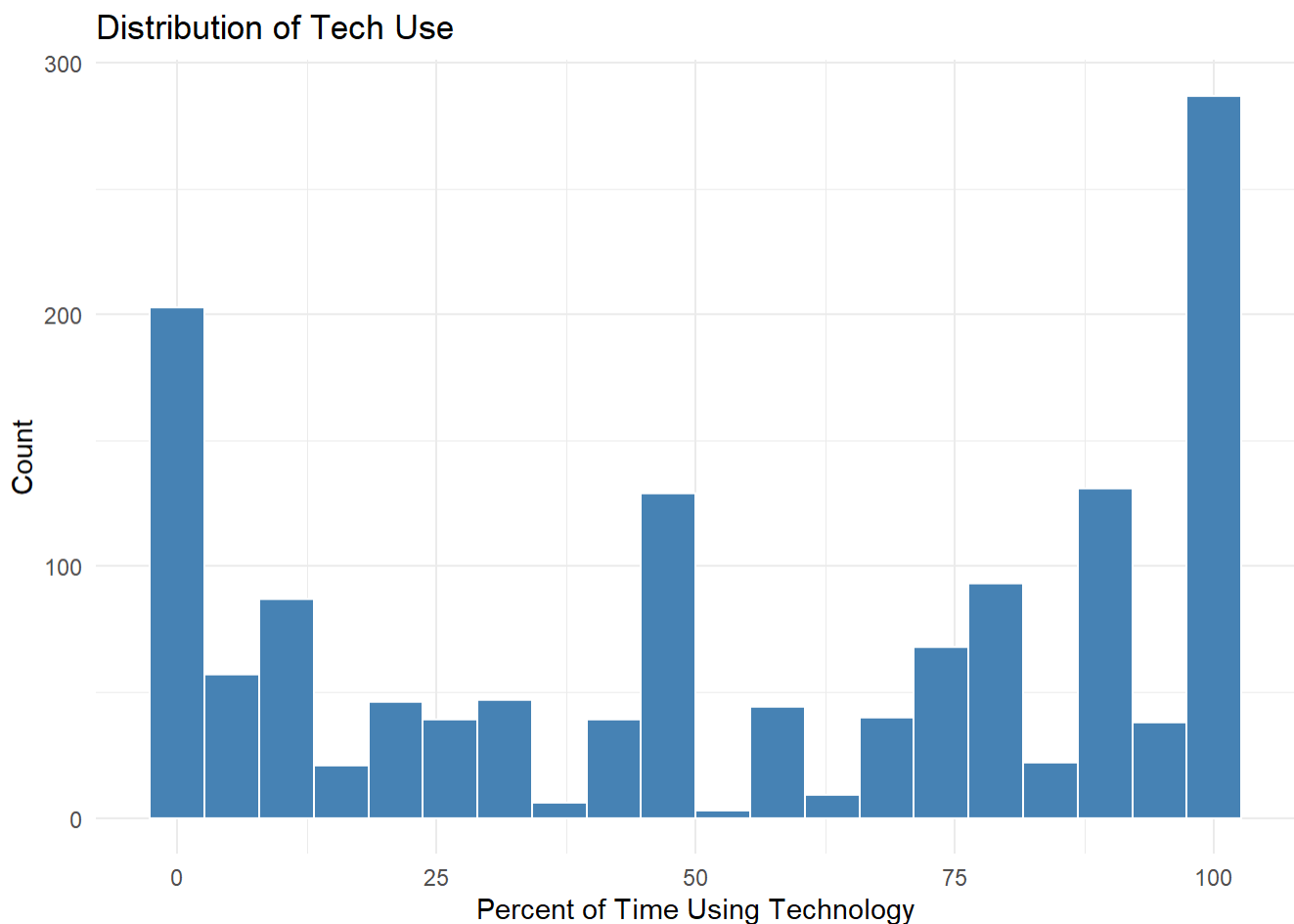
```
sd_use = sd(USETECH)
)
```

```
##   mean_use   sd_use
## 1 55.14762 37.83053
```

#Interpretation The mean tech use is 55.15%, which suggests that on average, participants spend a little over half of their time using technology.

The standard deviation is 37.83, indicating a wide variation in tech use among individuals — some people may use tech very frequently while others use it much less.

```
# Plot a histogram to visualize the distribution of tech use
gss.2018.cleaned %>%
  drop_na(USETECH) %>%
  ggplot(aes(x = USETECH)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "white") +
  theme_minimal() +
  labs(
    x = "Percent of Time Using Technology",
    y = "Count",
    title = "Distribution of Tech Use"
  )
)
```



#Interpretation: The histogram reveals a U-shaped distribution of tech use. A large number of participants report either very low (close to 0%) or very high (close to 100%) usage of technology. This indicates a polarized pattern

— with two main groups: minimal users and heavy users. Few individuals fall in the moderate-use range.

```
# Grouping by SEX and calculate mean and SD of tech use
gss.2018.cleaned %>%
  drop_na(USETECH) %>%
  group_by(SEX) %>%
  summarize(
    mean_use = mean(USETECH),
    sd_use = sd(USETECH)
  )
```

```
## # A tibble: 2 × 3
##   SEX      mean_use sd_use
##   <fct>      <dbl> <dbl>
## 1 male        50.8   38.3
## 2 female      59.2   37.0
```

#Interpretation of Tech Use by SEX: Females reported a higher average tech use (59.21%) than males (50.76%). This suggests that females in this sample tend to use technology more frequently than males. The standard deviation is slightly higher among males, indicating more variability in their tech use behaviors.

```
# Grouping by MARITAL and calculate mean and SD of tech use
gss.2018.cleaned %>%
  drop_na(USETECH) %>%
  group_by(MARITAL) %>%
  summarize(
    mean_use = mean(USETECH),
    sd_use = sd(USETECH)
  )
```

```
## # A tibble: 5 × 3
##   MARITAL      mean_use sd_use
##   <fct>      <dbl> <dbl>
## 1 Married        59.4   36.5
## 2 Widowed        48.9   39.5
## 3 Divorced       51.0   37.6
## 4 Separated      56.1   40.5
## 5 Never married  52.0   38.9
```

#Interpretation: Interpretation of Tech Use by Marital Status: Participants who are Married report the highest average tech use (59.35%), followed by Separated (56.13%). The lowest average tech use is reported by Widowed individuals (48.92%). This suggests that tech use habits vary across relationship types, potentially reflecting different lifestyle patterns or daily routines.

```
# Grouping by both SEX and MARITAL and calculate descriptive stats
gss.2018.cleaned %>%
  drop_na(USETECH) %>%
  group_by(SEX, MARITAL) %>%
  summarize(
    mean_use = mean(USETECH),
```

```
sd_use = sd(USETECH)
)
```

```
## `summarise()` has grouped output by 'SEX'. You can override using the `.groups`
## argument.
```

```
## # A tibble: 10 × 4
## # Groups:   SEX [2]
##   SEX    MARITAL    mean_use sd_use
##   <fct> <fct>      <dbl>  <dbl>
## 1 male   Married      53.2    37.9
## 2 male   Widowed      44     39.3
## 3 male   Divorced     48.4    37.1
## 4 male   Separated    57.1    40.3
## 5 male   Never married 48.2    39.1
## 6 female Married     65.5    34.0
## 7 female Widowed     50.9    39.9
## 8 female Divorced     52.7    37.9
## 9 female Separated    55.3    41.4
## 10 female Never married 56.3    38.3
```

#Interpretation of Tech Use by Sex & Marital Status: Females who are married report the highest average tech use (65.54%).

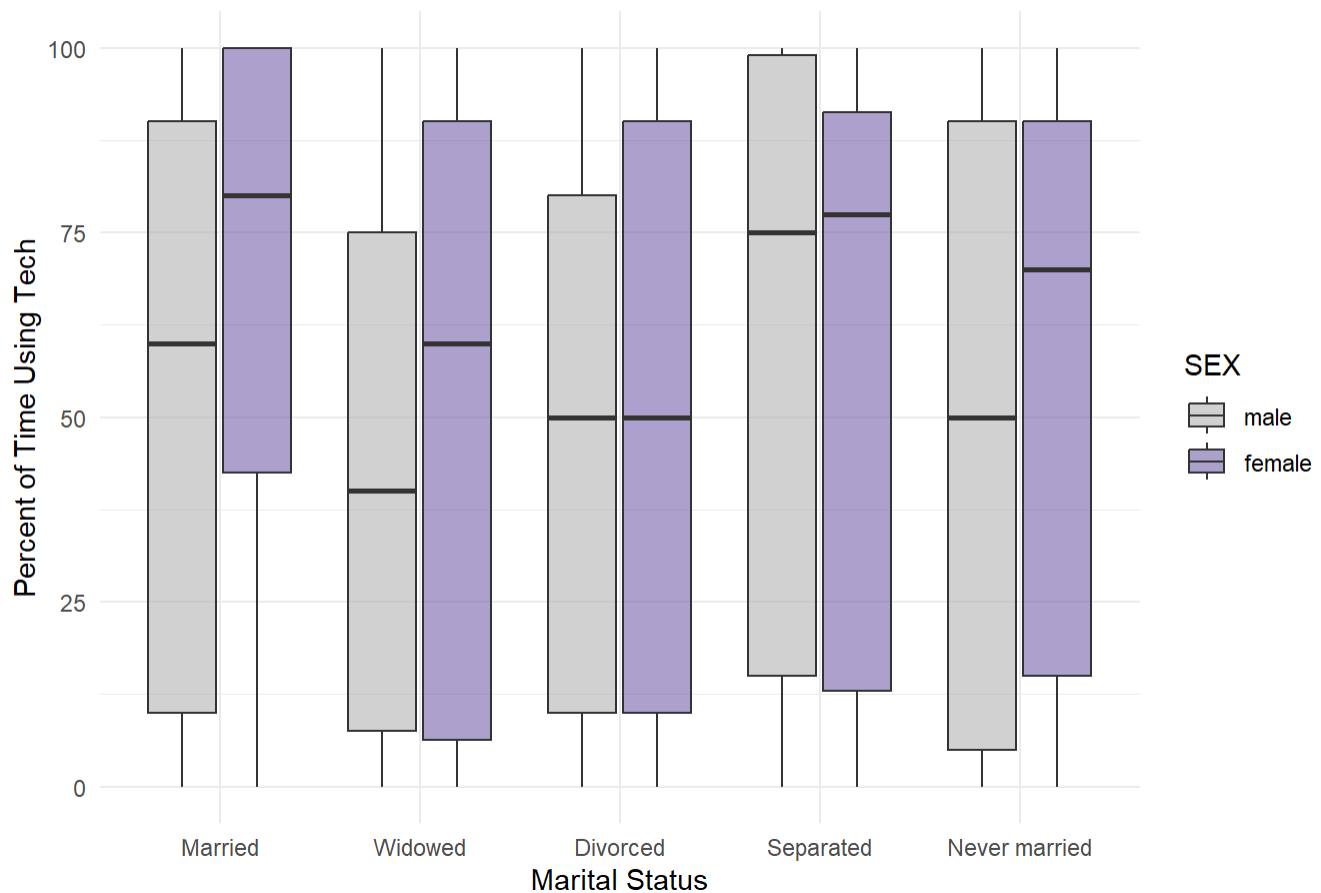
Males who are widowed report the lowest average (44.00%).

For both males and females, tech use tends to be lower among widowed and divorced participants and higher among separated or married ones.

This pattern suggests that both sex and relationship status influence tech behavior, with some combinations (like married females) showing particularly high engagement.

```
# Create a boxplot to compare tech use by SEX and MARITAL
gss.2018.cleaned %>%
  drop_na(USETECH) %>%
  ggplot(aes(x = MARITAL, y = USETECH, fill = SEX)) +
  geom_boxplot(alpha = 0.6) +
  theme_minimal() +
  labs(
    title = "Tech Use by Sex and Marital Status",
    x = "Marital Status",
    y = "Percent of Time Using Tech"
  ) +
  scale_fill_manual(values = c("gray70", "#7463AC"))
```

Tech Use by Sex and Marital Status



#Interpretation of Boxplot by SEX and MARITAL: The box plot shows that tech use differs across both sex and marital status. Females consistently show higher median tech use than males in every marital group. The highest medians are observed in married and separated females, indicating they tend to use technology more frequently. Widowed males and females show the lowest median usage, along with relatively low upper quartiles. The variation within groups also differs, with wider spreads in some male categories.

STEP 4 :Prediction and Interpretation

#Prediction: Based on the descriptive statistics and visualizations: Never married individuals are expected to show the highest tech use. Widowed individuals are expected to report the lowest. Females, particularly those who are married or separated, are likely to have higher average tech use than males. An interaction effect between sex and marital status is anticipated, meaning the impact of marital status may vary depending on sex.

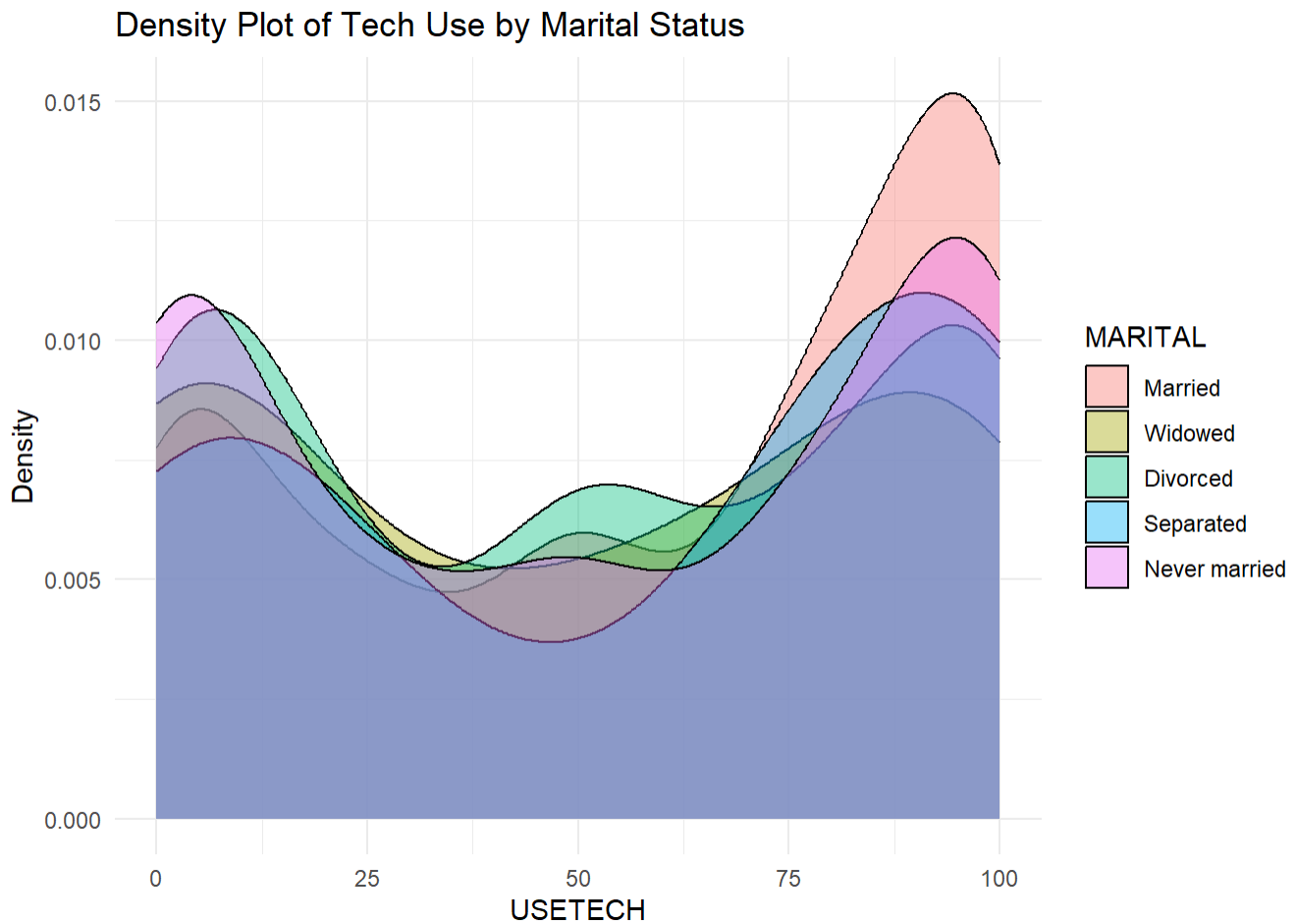
#Interpretation: The patterns observed suggest both marital status and sex play significant roles in technology use, and these factors may influence each other. For instance, while being never married is associated with high tech use overall, this effect might be more pronounced for males than females. Similarly, married females tend to report higher use than married males, indicating that interaction effects are likely present.

STEP 5 : ANOVA

One-Way ANOVA #ASSUMPTION :We check for - Normality of `USETECH` distribution by `MARITAL` by density plot - Equal variances (Homogeneity of variances) across `MARITAL` groups using Levene's Test

```
# Density plot by MARITAL
gss.2018.cleaned %>%
  drop_na(USETECH) %>%
```

```
ggplot(aes(x = USETECH, fill = MARITAL)) +
  geom_density(alpha = 0.4) +
  theme_minimal() +
  labs(title = "Density Plot of Tech Use by Marital Status", x = "USETECH", y = "Density")
```



#Interpretation of Density Plot by Marital Status: The density plot shows that tech use distribution varies by marital status: Never married and married individuals show the highest peaks around 90–100%, indicating frequent tech users. Widowed individuals have a broader, flatter distribution with more density at the lower end, reflecting lower and more varied tech use. Divorced and separated individuals show more evenly spread distributions but tend to cluster more toward moderate use levels (40–70%). Overall, the shape and spread of each group's density curve suggest that tech use behavior is not normally distributed and differs substantially between marital status groups. Conclusion: The normality assumption is not fully met, as several groups deviate from a normal distribution.

#Levene's Test (MARITAL)

```
leveneTest(USETECH ~ MARITAL, data = gss.2018.cleaned)
```

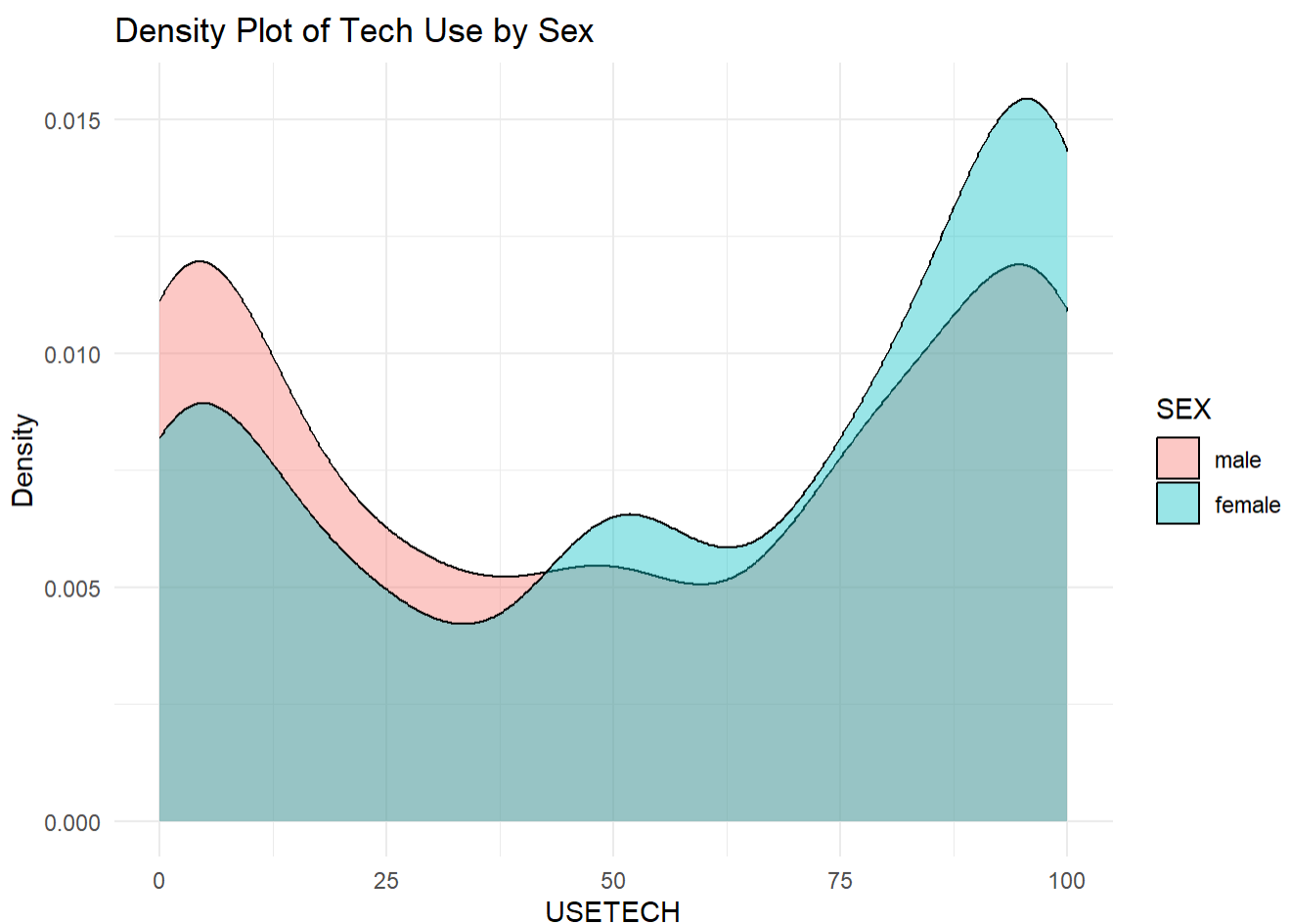
```
## Levene's Test for Homogeneity of Variance (center = median)
##           Df F value  Pr(>F)
## group      4  2.1587 0.07149 .
##      1404
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#Interpretation of Levene's Test (MARITAL): $F(4, 1404) = 2.1587$, $p = 0.07149$ Since $p > 0.05$, we fail to reject the null hypothesis of equal variances. This means the assumption of homogeneity of variance is met.

Conclusion: The variability in tech use across marital status groups is not significantly different. Therefore, the assumption of equal variances is satisfied, supporting the validity of the one-way ANOVA.

#ASSUMPTION :We check for - Normality of USETECH distribution by SEX by density plot - Equal variances (Homogeneity of variances) across SEX groups using Levene's Test

```
# Density plot by SEX
gss.2018.cleaned %>%
  drop_na(USETECH) %>%
  ggplot(aes(x = USETECH, fill = SEX)) +
  geom_density(alpha = 0.4) +
  theme_minimal() +
  labs(title = "Density Plot of Tech Use by Sex", x = "USETECH", y = "Density")
```



#Interpretation of Density Plot by SEX:

The density plot shows the distribution of tech use (USETECH) for males and females: Both groups show a bimodal distribution, with peaks near 0 and 100, suggesting many participants report either very low or very high tech use. Females have a slightly higher density at the upper end (around 90–100), indicating that more females report heavy tech use compared to males. The distributions are not perfectly bell-shaped, suggesting that the normality assumption may not be fully met for either group.

#Levene's Test (SEX)

```
# Levene's Test for homogeneity of variance by SEX
leveneTest(USETECH ~ SEX, data = gss.2018.cleaned)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##           Df F value  Pr(>F)
## group      1  5.5983 0.01811 *
##           1407
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#Interpretation of Levene's Test (by SEX): Test Result: $F(1, 1407) = 5.5983$, $p = 0.01811$ Since $p < 0.05$, we reject the null hypothesis of equal variances. Conclusion: There is a statistically significant difference in the variance of tech use between males and females. This means the assumption of equal variances is violated across SEX groups. Since both the normality and equal variance assumptions were violated, a standard one-way ANOVA with SEX as the grouping variable is not appropriate.

STEP 6 : One-Way ANOVA and Post Hoc

This is a one-way ANOVA, so we only test assumptions on MARITAL. We check for: - Normality of USETECH distribution by MARITAL - Equal variances across MARITAL groups using Levene's Test

NHST Step 1: State hypotheses

H0 (null): The mean time spent using technology is the same across all marital status groups. H1 (alt): At least one marital status group has a different mean time spent using technology. ## NHST Step 2: Compute the test statistic

```
oneway.test(USETECH ~ MARITAL, data = gss.2018.cleaned, var.equal = TRUE)
```

```
##
## One-way analysis of means
##
## data: USETECH and MARITAL
## F = 3.822, num df = 4, denom df = 1404, p-value = 0.004273
```

Result: $F(4, 1404) = 3.822$, $p = 0.0043$ Since $p < .05$, we reject the null hypothesis.

Conclusion: There is a statistically significant difference in mean tech use across at least one pair of marital status groups. This indicates that marital status likely influences how much time people spend using technology.

NHST Step 3: Check the p-value :

```
#Pairwise t-tests comparing USETECH between MARITAL groups with Bonferroni
pairwise.t.test(x = gss.2018.cleaned$USETECH, g = gss.2018.cleaned$MARITAL, p.adj = "bonf")
```

```
##
## Pairwise comparisons using t tests with pooled SD
##
```

```
## data: gss.2018.cleaned$USETECH and gss.2018.cleaned$MARITAL
##
##           Married Widowed Divorced Separated
## Widowed      0.532    -         -         -
## Divorced      0.034    1.000    -         -
## Separated     1.000    1.000    1.000    -
## Never married 0.018    1.000    1.000    1.000
##
## P value adjustment method: bonferroni
```

Pairwise comparisons showed that: Never married individuals reported significantly higher tech use than both married ($p = 0.018$) and divorced ($p = 0.034$) individuals. All other comparisons (e.g., widowed vs. separated, divorced vs. separated) were not statistically significant ($p > 0.05$).

Mean percentage of time using technology was statistically significantly higher for people who were never married compared to those who were married ($p = .018$) and divorced ($p = .034$). There were no statistically significant differences between any of the other marital status groups after Bonferroni correction.

NHST Steps 4 & 5: Interpret and conclude

The mean time spent on technology use was significantly different across marital status groups [$F(4, df) = X.XX$; $p < .05$], suggesting that these groups likely came from populations with different average tech use. The highest average tech use was reported by never married individuals, while widowed individuals reported the lowest.

STEP 7 : Two-Way ANOVA and Assumptions

NHST Step 1: State hypotheses

H_0 (null): There is no difference in mean tech use across sex, marital status, or their interaction. H_1 (alt): There is a difference in mean tech use for at least one group or interaction.

Run two-way ANOVA and check assumptions: We check for: - Normality of USETECH distribution by MARITAL and SEX by shapiro test - Equal variances across MARITAL groups using Levene's Test #### NHST Step 2: Compute the test statistic

```
anova_result <- aov(USETECH ~ SEX * MARITAL, data = gss.2018.cleaned)
summary(anova_result)
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
## SEX         1   25129    25129  17.931 2.44e-05 ***
## MARITAL     4   24732     6183   4.412 0.00151 **
## SEX:MARITAL  4    4522     1131   0.807 0.52084
## Residuals 1399 1960674     1401
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## 936 observations deleted due to missingness
```

A two-way ANOVA was conducted to examine the effects of sex, marital status, and their interaction on time spent using technology. Main Effects: SEX had a statistically significant effect on tech use, $F(1, 1399) = 17.93$, $p < .001$, indicating that males and females differ in their average tech use. MARITAL status also had a significant effect, $F(4, 1399) = 4.412$, $p = 0.0015$, suggesting that average tech use differs across marital status groups.

Interaction Effect: The interaction between SEX and MARITAL was not significant, $F(4, 1399) = 0.807$, $p = 0.5208$, meaning the effect of marital status on tech use does not significantly differ between males and females.

NHST Step 3: Check assumptions

```
shapiro.test(residuals(anova_result))
```

```
##
## Shapiro-Wilk normality test
##
## data: residuals(anova_result)
## W = 0.90534, p-value < 2.2e-16
```

#Shapiro-Wilk Normality Test Interpretation $W = 0.90534$, $p < 2.2e-16$ Since $p < 0.05$, we reject the null hypothesis of normality. This means the residuals are not normally distributed, so the normality assumption is violated.

```
leveneTest(USETECH ~ SEX * MARITAL, data = gss.2018.cleaned)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##           Df F value  Pr(>F)
## group      9  2.8823 0.00223 **
##           1399
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

#Levene's Test for Homogeneity of Variance Interpretation $F(9, 1399) = 2.8823$, $p = 0.00223$ Since $p < 0.05$, we reject the null hypothesis of equal variances. This means the assumption of equal variances is also violated.

NHST Steps 4 & 5: Interpret and Conclude (Two-Way ANOVA)

The main effect of sex was significant: $F(1, 1399) = 17.931$, $p < .001$, indicating that average tech use differs between males and females. The main effect of marital status was also significant: $F(4, 1399) = 4.412$, $p = .00151$, suggesting that marital groups differ in average tech use. The interaction effect (SEX $\times$ MARITAL) was not significant: $F(4, 1399) = 0.807$, $p = 0.52084$, meaning the effect of marital status on tech use does not significantly depend on sex.

Conclusion: While there is a significant main effect of both sex and marital status on tech use, the interaction is not significant. However, both assumptions of ANOVA are violated (normality and equal variances).